

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 4

Time Duration : 1 Hour
Total: Marks:40

Code of Conduct:

*I Shaik. Abdul Razak (192524358) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Shaik. Abdul Razak

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Task 1: Data Preparation and Inductive Learning

(a) Inductive Learning Approach

Inductive learning means learning a general pattern from previously available examples and then using that pattern to predict the result for a new patient.

In this problem, the previous patient records contain information such as age, blood pressure, cholesterol, glucose, BMI and family history. The model learns the relationship between these factors and whether the patient has diabetes.

The **target variable** is:

Diabetes Status → Diabetic / Non-Diabetic

Some important input features are:

1. **Glucose Level:** This is an important indicator because abnormal glucose levels are strongly related to diabetes.
2. **BMI:** A higher BMI can indicate increased risk of developing type-2 diabetes.
3. **Age:** Diabetes risk can increase with age.
4. **Blood Pressure:** High blood pressure can be associated with metabolic health problems.
5. **Family History:** A family history of diabetes can indicate increased genetic risk.

For this dataset, I would start with a **Decision Tree classifier** because it can identify relationships between these features and diabetes in an easy-to-understand form.

(b) Handling Class Imbalance

Suppose the dataset contains:

- 800 non-diabetic patients
- 200 diabetic patients

Here, the diabetic class is the minority class. If the model is trained directly, it may become biased toward predicting patients as non-diabetic.

I would use **SMOTE** to address this problem.

SMOTE creates new synthetic samples for the minority class using existing minority-class observations. This gives the model more examples of diabetic patients during training.

I prefer SMOTE over under-sampling because under-sampling would remove some non-diabetic records, which means useful information could be lost.

One important point is that SMOTE should be applied **after the train-test split and only to the training data**. Otherwise, information from the test data could leak into the training process.

For a medical application, I would particularly monitor **Recall**, because missing an actual diabetic patient is more serious than incorrectly flagging some non-diabetic patients.

(c) Data Split and Preprocessing

I would divide the dataset into:

70% Training Data + 30% Testing Data

The training portion is used to learn the patterns, while the testing portion is kept separate to check how well the model works on unseen patients.

I would use a **stratified split** so that the diabetic/non-diabetic ratio remains approximately similar in both datasets.

Preprocessing steps include:

1. Check for missing values.
2. Replace missing numerical values using median values where appropriate.
3. Convert categorical information such as family history into numerical form.
4. Standardise numerical features for Logistic Regression.
5. Apply SMOTE only to the training data.
6. Keep the test data unchanged.

Normalisation is especially useful for statistical models because features such as age, cholesterol and glucose have different numerical ranges. Decision Trees normally do not require feature scaling.

Task 2: Decision Tree for Diabetes Diagnosis

(a) Decision Tree Construction

I would train a Decision Tree using the training dataset. The tree chooses a feature that gives the best separation between diabetic and non-diabetic patients.

Two commonly used measures are:

Gini Index

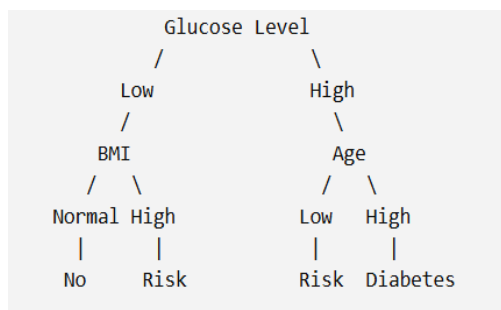
A lower Gini value represents a purer group.

Information Gain

The feature having the highest Information Gain can be selected as the root.

For this particular medical problem, **glucose level is expected to be one of the strongest candidates for the root node**, because it provides strong information about diabetes status.

An example of the first three levels is:



The above is an example for explaining the structure. The actual tree should be generated from the available dataset.

(b) Model Evaluation

After training, I would calculate four major measures.

Accuracy

Precision

Recall

F1 Score

The results can be reported in the following format:

Measure	Decision Tree
Accuracy	Obtained from test data
Precision	Obtained from test data
Recall	Obtained from test data
F1-Score	Obtained from test data

A **False Negative** occurs when:

Actual patient = Diabetic

Model prediction = Non-Diabetic

This is an important problem in healthcare because the patient may not receive additional testing or treatment at the correct time.

To reduce false negatives, I would:

- Give more importance to the diabetic class.
- Use SMOTE.
- Adjust the decision threshold if appropriate.
- Focus on improving recall.
- Use clinical validation before deployment.

In this situation, accepting some additional false positives can be reasonable if it helps identify more diabetic patients for further medical examination.

(c) Pre-Pruning and Post Pruning

A Decision Tree can become very large if it is allowed to grow without restrictions. Such a tree can memorise the training data instead of learning general patterns. This is called **overfitting**.

Pre-pruning

Pre-pruning controls the tree while it is being created.

Examples include:

- Maximum tree depth
- Minimum samples required for splitting
- Minimum samples in a leaf

Post-pruning

Post-pruning first creates a larger tree and then removes branches that do not provide enough useful information.

Cost-complexity pruning uses:

where controls the penalty for tree complexity.

A comparison can be presented as:

Model	Main Advantage	Main Problem
Pre-pruned Tree	Simple and fast	May remove useful patterns
Fully grown Tree	Learns detailed patterns	Can overfit
Post-pruned Tree	Reduces unnecessary branches	Requires choosing pruning parameter

For deployment in a clinical environment, I would prefer a **pruned Decision Tree** if it provides good recall and test performance because it is easier to understand and less likely to overfit

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

I would also apply **Logistic Regression** to the same dataset.

Logistic Regression predicts the probability that a patient belongs to the diabetic class.

The basic equation is:

where:

The model can produce a value such as:

Probability of diabetes = 0.78

A threshold can then be used to convert the probability into a class.

The models can be compared using:

Model	Accuracy	AUC-ROC
Decision Tree	Calculated value	Calculated value
Logistic Regression	Calculated value	Calculated value

AUC-ROC is useful because it shows how well the model separates diabetic and non-diabetic patients over different classification thresholds.

Logistic Regression is a good choice when:

- The dataset is not extremely complex.
- A probability/risk score is required.
- The relationship between features and diabetes is reasonably systematic.
- We need a relatively simple model.

(b) Feature Importance

I would examine the important features from both models.

For the Decision Tree, importance can be calculated from the reduction in impurity caused by each feature.

For Logistic Regression, the coefficients can be examined. After appropriate scaling, larger absolute coefficients indicate stronger influence on the predicted outcome.

An example result could be:

Rank	Decision Tree	Logistic Regression
1	Glucose	Glucose
2	BMI	BMI
3	Age	Family History

The exact ranking depends on the actual patient dataset.

If both models identify glucose as the most important feature, it indicates that glucose contains strong predictive information. The models may disagree about other features because they learn relationships differently.

A Decision Tree can identify threshold-based relationships, while Logistic Regression mainly models the effect of features through a mathematical relationship with the log-odds of diabetes.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

The treatment recommendation problem can be represented using a **Markov Decision Process**.

An MDP consists mainly of:

States + Actions + Rewards + Transitions

States

The patient's condition can be divided into different stages:

State	Description
S0	Normal/Low Risk
S1	Moderate Risk
S2	High Risk
S3	Diabetic

The state can be decided using patient measurements such as glucose, BMI and blood pressure.

Actions

The agent has four possible actions:

- **Diet**
- **Exercise**
- **Medication**
- **Monitor**

Reward

The agent should receive a positive reward when the patient's health condition improves.

Example:

Situation	Reward
Health improves	+10
Health remains stable	+3
Health becomes worse	-10
Unsafe action	-20

These numbers are only an example. In a real healthcare system, the reward function should be designed with medical experts.

Transition

A transition describes how the patient's condition changes after an action.

For example:

Moderate Risk

|

Exercise

↓

Improved State

Another possible transition is:

High Risk

|

Monitor

↓

No Improvement

↓

High Risk

The probabilities of these transitions can be estimated from historical patient data or carefully designed simulations.

(b) Q-Learning

Q-Learning learns which action gives the highest long-term reward for each patient state.

The update formula is:

where:

- s = current state
- a = selected action
- R = received reward
- S' = next state
- α = learning rate
- γ = discount factor

For example, let:

Suppose:

Then:

Two iterations

Iteration	State	Action	Old Q	Reward	New Q
1	Moderate Risk	Exercise	2.00	4	2.47
1	High Risk	Medication	1.50	5	1.98
1	Low Risk	Diet	1.00	3	1.37
2	Moderate Risk	Exercise	2.47	3	2.83
2	High Risk	Medication	1.98	4	2.54
2	Low Risk	Diet	1.37	2	1.61

The agent should be trained for **at least five patient episodes**. During each episode, the agent observes the state, chooses an action, receives a reward and moves to another state.

Convergence

I would consider the Q-learning process stable when the change between consecutive Q-tables becomes very small:

for several consecutive episodes.

(c) Final Learned Policy and Ethics

The final policy is obtained by selecting the action with the highest Q-value:

An example learned policy could be:

Patient Condition	Selected Action
Low Risk	Exercise
Moderate Risk	Diet
High Risk	Medication*
Diabetic	Medication*
Stable/Improved	Monitor

*Medication decisions must **not** be made autonomously by an RL system in a real clinical environment.

Ethical Issues

Patient Safety:

A wrong recommendation can cause harm. Therefore, the system should have safety restrictions and clinician approval.

Bias:

If the training data contains unequal representation of different patient groups, the learned policy may produce unfair recommendations.

Explainability:

Doctors should be able to understand why the system selected a particular action.

Human Control:

The AI should support doctors rather than completely replace their decisions.

Privacy:

Patient records contain sensitive information and must be securely stored and processed.

Testing:

The system should be extensively tested before being used with real patients.

Therefore, This system combines three machine-learning approaches for different purposes.

Decision Tree and Logistic Regression can be used to predict diabetes and estimate patient risk from historical records. **Q-Learning** can then be used as a research framework for learning treatment strategies over changing patient states. Among the diagnostic models, the final choice should be based not only on accuracy but also on recall, AUC, interpretability and reliability. For treatment recommendation, the RL system should always work as a **decision-support tool under medical supervision**, rather than independently deciding patient treatment.